SENTIMENT ANALYSIS DASHBOARD

FOR SOCIAL MEDIA

A simple way to understand what people feel and say online

Project Presentation • BCA / Data Science

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1. Introduction

- Social media platforms contain millions of posts, comments, reviews, and replies every day.
- People express positive, negative, or neutral opinions about products, services, events, and public topics.
- Sentiment analysis uses Natural Language Processing (NLP) to identify the opinion or emotion in text.
- A dashboard displays the results using charts, summary cards, filters, and recent posts.

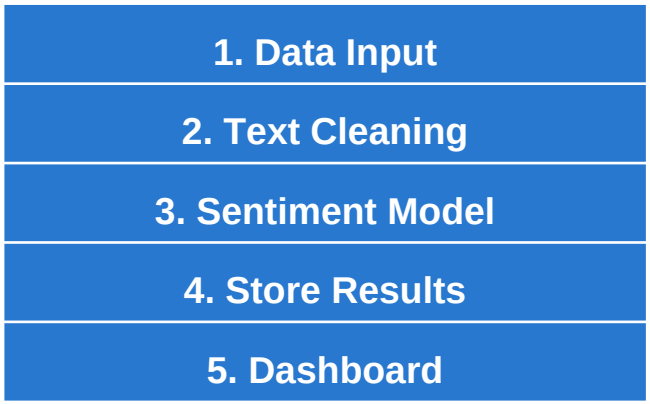
2. Problem Statement

- Reading a large number of social media comments manually takes time.
- It is difficult to quickly identify whether public opinion is positive, negative, or neutral.
- Important changes in customer satisfaction or public reaction may be missed.
- The project provides a single visual dashboard to summarize text sentiment and trends.

3. Objectives

- Collect or upload social media text such as posts, comments, and reviews.
- Clean text by handling links, repeated spaces, and unwanted symbols.
- Classify each text as positive, negative, or neutral.
- Show sentiment percentages, post counts, keywords, and changes over time.
- Help users explore results using date, platform, or keyword filters.

4. System Workflow



Output:

- Overall sentiment summary
- Positive / negative / neutral split
- Trend over time
- Frequent words and example posts

Typical tools:

Python, pandas, NLP library, Streamlit, Plotly

5. Main Dashboard Features

- Summary cards: total posts analyzed and positive, negative, and neutral percentages.
- Sentiment chart: compares the number of posts in each sentiment category.
- Trend chart: shows how sentiment changes by day or week.
- Keyword view: highlights common words or topics in the collected text.
- Search and filters: narrow results by platform, date, keyword, or sentiment.
- Post table: displays text, predicted sentiment, and confidence score when available.

6. Sample Dashboard (Illustrative Data)

1,000	55%	25%	20%
Posts analyzed	Positive	Negative	Neutral

Note: The numbers above are sample values for demonstration, not live social media data.

7. Technologies Used

- Python: main programming language for data processing and analysis.
- pandas: organizes and cleans text data in tables.
- NLTK, VADER, or a transformer model: analyzes sentiment in text.
- Streamlit: builds an interactive dashboard with Python.
- Plotly or Matplotlib: creates charts and graphs.
- CSV or a database: stores collected posts and analysis results.

8. Example Use Cases

- Brand monitoring: understand reactions to a product launch or campaign.
- Customer service: identify negative feedback that may need attention.
- Event analysis: track public reactions to an event or announcement.
- Market research: compare opinions about products or services.
- Community management: observe topics that are receiving positive or negative responses.

9. Benefits and Limitations

- Benefits: saves manual effort, summarizes large text collections, and reveals trends quickly.
- Benefits: visual charts make results easier to understand and share.
- Limitations: sarcasm, slang, emojis, mixed languages, and context can cause incorrect predictions.
- Limitations: results depend on the quality and representativeness of the collected data.
- Privacy: use permitted data, protect user information, and follow platform rules.

10. Future Enhancements

- Connect approved platform APIs for regularly updated data.
- Support multiple languages and regional language text.
- Add topic detection to identify what people are discussing.
- Send alerts when negative sentiment increases sharply.
- Evaluate model accuracy using labeled examples and improve it over time.

11. Conclusion

- A sentiment analysis dashboard turns social media text into understandable summaries.
- It combines text preprocessing, sentiment classification, and data visualization.
- The dashboard can support better decisions in marketing, customer service, and research.
- Its results should be treated as predictions, not perfect measurements of public opinion.

12. References

- Python documentation: https://docs.python.org/3/
- pandas documentation: https://pandas.pydata.org/docs/
- Streamlit documentation: https://docs.streamlit.io/
- NLTK documentation: https://www.nltk.org/
- Plotly documentation: https://plotly.com/python/

Thank you! • Questions?